

Anomaly Detection in District Heating Substations Using Reconstruction-Based Models

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Anomaly Detection in District Heating Substations Using Reconstruction-Based Models

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This thesis investigates anomaly detection in a small district-heating network using historical operational data from multiple consumer substations. The work starts from the HEAT paper as a methodological reference, but adapts the approach to a setting with fewer substations, longer historical coverage, and limited fault labels. The main method is a reconstruction-based anomaly detector that models 24-hour windows of supply temperature, return temperature, and derived flow. The study compares engineering baselines, thresholding rules, dominant-feature attribution, and clustering-based interpretation in order to identify anomaly candidates that can be reviewed with domain experts and later applied to live operational data.

SAMMANFATTNING

Detta examensarbete undersöker anomalidetektion i ett litet fjärrvärmenät med hjälp av historiska driftsdata från flera kundcentraler. Arbetet utgår från HEAT-artikeln som metodologisk referens, men anpassar angreppssättet till en situation med färre undercentraler, längre historisk täckning och begränsad tillgång till felmärkta data. Huvudmetoden är en rekonstruktionsbaserad anomalidetektor som modellerar 24-timmarsfönster av framledningstemperatur, returtemperatur och härlett flöde. Studien jämför ingenjörsmässiga baslinjer, tröskelvärdesmetoder, dominant-funktionsattribuering och klusterbaserad tolkning för att identifiera anomalikandidater som kan granskas tillsammans med domänexperter och senare användas på live-data.

Keywords: district heating, anomaly detection, autoencoder, reconstruction error, time-series analysis

Nyckelord: fjärrvärme, anomalidetektion, autoencoder, rekonstruktionsfel, tidsserieanalys

CONTENTS

Abstract	1
Sammanfattning	1
Contents	1
1 Introduction	3
2 Background	3
2.1 District-heating anomaly detection	3
2.2 HEAT paper as reference	3
2.3 Reconstruction-based anomaly detection	4
2.4 Physics-informed feature construction	4
3 Data	4
3.1 Available sources	4

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3.2	Main measured variables	5
3.3	Preprocessing decisions	5
3.4	Known limitations	5
3.5	Five-building thesis scope	5
4	Method	6
4.1	Overview	6
4.2	Autoencoder architecture	6
4.3	Feature construction	6
4.4	Thresholding	6
4.5	Engineering baseline	6
4.6	Clustering roles	7
5	Results	8
5.1	Current main result set	8
5.2	Dominant-feature interpretation	9
5.3	Low delta-T overlap	9
5.4	Operating regimes	10
5.5	Threshold comparison	10
5.6	Flow-channel sensitivity comparison	11
5.7	Train-test split and anomaly timing	11
6	Discussion	11
6.1	What is currently convincing	11
6.2	What remains uncertain	11
6.3	Implications for live deployment	12
6.4	Writing priorities	12
7	Conclusion	12
	Acknowledgments	12
	References	12
A	Writing notes	12

1 INTRODUCTION

District-heating substations operate continuously and generate large volumes of time-series data, but real fault labels are often sparse or unavailable. This creates a practical detection problem: operators still need a way to identify unusual operating periods, even when a supervised classification setup is not possible. The aim of this thesis is therefore to build and evaluate an anomaly-detection pipeline that can identify physically interpretable abnormal behavior in district-heating substations using historical data and later be applied to live data.

The work is motivated by the HEAT paper, which combines encoder-based representation learning and clustering for fault detection in district-heating substations. However, the available KTH thesis dataset differs in two important ways. First, the network is much smaller, which weakens the statistical value of peer-group clustering. Second, the historical coverage is much longer, which makes per-substation temporal modeling more promising than network-wide peer comparison alone. Because of this, the thesis focuses primarily on reconstruction-based anomaly detection and treats clustering as a supporting interpretation tool rather than the central detection mechanism.

The main research question is:

Can reconstruction-based anomaly detection on supply temperature, return temperature, and derived flow identify interpretable abnormal operating periods in a small district-heating network?

The thesis also addresses the following supporting questions:

- How should the derived flow feature be handled when small temperature differences cause numerical instability?
- How does a reconstruction-based detector compare with a simple engineering low delta-T baseline?
- What is the effect of threshold choice on the number and type of detected anomalies?
- Can clustering still help as an interpretation layer when the number of substations is small?

The rest of the thesis is organized as follows. Chapter 2 reviews the relevant background and the HEAT paper. Chapter 3 describes the available datasets and preprocessing assumptions. Chapter 4 presents the anomaly-detection methodology. Chapter 5 summarizes the experimental results. Chapter 6 discusses interpretation, limitations, and implications for live deployment. Chapter 7 concludes the thesis and outlines future work.

2 BACKGROUND

2.1 District-heating anomaly detection

District-heating systems are monitored through temperature, flow, and power-related signals that reflect both thermal performance and hydraulic behavior. In practice, low supply-return temperature difference is often used as a first engineering warning sign because it may indicate poor heat extraction, inefficient operation, or high return-temperature behavior. However, low delta-T only captures one narrow anomaly type and cannot describe all abnormal operating patterns.

2.2 HEAT paper as reference

The HEAT paper proposes a hierarchical-constrained, encoder-assisted clustering approach for fault detection in district-heating substations. Its main idea is to learn compact representations of time-series

behavior and then use clustering to approximate peer groups for local comparison. This is well suited to a large network with many substations, where peer comparison can be statistically meaningful.

In this thesis, HEAT remains the main methodological reference. The adaptation is necessary because the available dataset has far fewer consumer sheets than the HEAT case study. The practical consequence is that clustering is still valuable, but its role shifts from being the primary fault-detection structure to being a secondary interpretation layer.

2.3 Reconstruction-based anomaly detection

An autoencoder is a neural network trained to reconstruct its own input. In the thesis pipeline, each input is a 24-hour window containing three channels: supply temperature, return temperature, and derived flow. The encoder compresses the window into a latent representation, and the decoder reconstructs the original signal. If a window is typical of historical operating behavior, the reconstruction error is low. If a window is unusual, the reconstruction error tends to increase. This provides a natural anomaly score without requiring explicit labels.

2.4 Physics-informed feature construction

The third input channel is derived from the thermal balance equation:

$$\dot{m} = \frac{P}{c_p(T_s - T_r)}$$

where P is thermal power, c_p is the specific heat capacity of water, T_s is supply temperature, and T_r is return temperature. In this thesis, the power measurement is treated as kilowatts for all sheets based on supervisor guidance. A practical difficulty is that very small $T_s - T_r$ can cause the derived flow to become numerically unstable. For that reason, both a raw-flow and a stabilized-flow formulation are retained in the analysis, with stabilized flow used as the main reporting lens.

3 DATA

3.1 Available sources

The thesis uses historical data from a district-heating network associated with Montserrat / Abat Oliba. The broader exploratory phase reviewed seven heating-consumer sheets. For the current draft and the main thesis narrative, the analysis is narrowed to the five sheets with the most usable data and without long faulty constant-value periods:

- cons_abat_cisneros
- cons_abat_garriga
- cons_abat_marcet
- cons_abat_oliba
- cons_hostatgeria_underfloor_heating

Two additional sheets, cons_hostatgeria_DHW_radiators and cons_nostra_senyora, were retained during the exploratory phase but are not used in the main thesis draft because they contain long periods of faulty constant-value behavior that make interpretation less reliable.

These sheets are complemented by reports, drawings, and contextual material about the network. This contextual material is valuable because it helps interpret anomaly candidates even when explicit fault labels are not available.

3.2 Main measured variables

The most consistently available variables across the modeled heating-consumer sheets are:

- timestamp
- supply temperature
- return temperature
- power

Derived features are then computed from these measured variables, most notably delta-T and derived flow.

3.3 Preprocessing decisions

The current preprocessing pipeline uses the following decisions:

- active-heating rows are selected using positive power and positive delta-T
- data is resampled to 15-minute intervals
- windows are formed as 24-hour segments with 12-hour stride
- windows with low data completeness or insufficient active-heating share are discarded
- channels are normalized before being passed to the autoencoder

These decisions are intended to focus the model on meaningful heating behavior rather than long inactive periods.

3.4 Known limitations

The most important limitations of the dataset are:

- lack of reliable fault labels
- heterogeneity across substations
- possible instability in derived flow when delta-T is small
- variation in data quality, including zeros and missing periods

Because of these limitations, evaluation in this thesis is based on anomaly review, cross-method comparison, and supervisor/domain interpretation rather than standard supervised metrics.

3.5 Five-building thesis scope

The results chapter therefore focuses on the following five-building subset:

- Abat Cisneros
- Abat Garriga
- Abat Marcet
- Abat Oliba
- Hostatgeria Underfloor

This choice is practical rather than theoretical. The discarded sheets are not treated as evidence against the method; they are simply weaker as thesis cases because faulty constant-value periods introduce ambiguity that is difficult to resolve without stronger data validation.

4 METHOD

4.1 Overview

The final workflow used in the thesis is:

- (1) preprocess each heating-consumer sheet into retained 24-hour windows
- (2) construct three model channels: supply temperature, return temperature, and a processed flow channel
- (3) train a 1D convolutional autoencoder on the historical windows
- (4) compute reconstruction error for each window
- (5) flag anomalous windows using a training-distribution threshold
- (6) attribute each anomaly to its dominant feature using per-feature reconstruction error
- (7) compare the detected anomalies with engineering baselines and clustering-based interpretation views

4.2 Autoencoder architecture

The current architecture is a compact 1D convolutional autoencoder. The encoder contains two convolutional blocks with ReLU activations and max pooling, while the decoder reconstructs the signal using transposed convolutions. The model is intentionally conservative and was used as a first reliable baseline for reconstruction-based anomaly detection.

4.3 Feature construction

The model uses three synchronized channels per 24-hour window:

- supply temperature
- return temperature
- flow channel

The flow channel is included because thermal anomalies are often easier to interpret when temperature behavior is viewed together with flow-related behavior. During method development, several formulations of the flow signal were explored in order to improve numerical robustness. In the final draft, the emphasis is therefore on the behavior of the input flow channel rather than on the intermediate engineering details of every tested formulation.

4.4 Thresholding

Two threshold rules were evaluated:

- training 99th percentile (p_{99})
- training mean plus three standard deviations (3-sigma)

The latest main result pack uses the 3-sigma rule, following supervisor guidance. The thesis should discuss that neither threshold is uniformly stricter across all sheets; the difference depends on the shape of the training reconstruction-error distribution.

4.5 Engineering baseline

The engineering baseline is a low delta-T anomaly rule computed only on active-heating rows. It is not used by the autoencoder. Instead, it is used afterward to check whether detected anomaly windows overlap a simple, physically interpretable warning signal.

Autoencoder concept used in this project

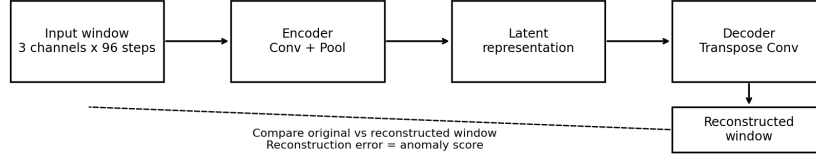


Figure 1: Conceptual view of the reconstruction-based anomaly detector. A 24-hour window containing supply temperature, return temperature, and the flow channel is encoded, reconstructed, and scored by reconstruction error.

4.6 Clustering roles

Two clustering roles were explored:

- operating-regime clustering on all windows, using summary statistics of each day
- anomaly-only clustering on flagged windows, used as a secondary anomaly-family interpretation layer

The thesis should make clear that clustering is supportive rather than central in the final pipeline.

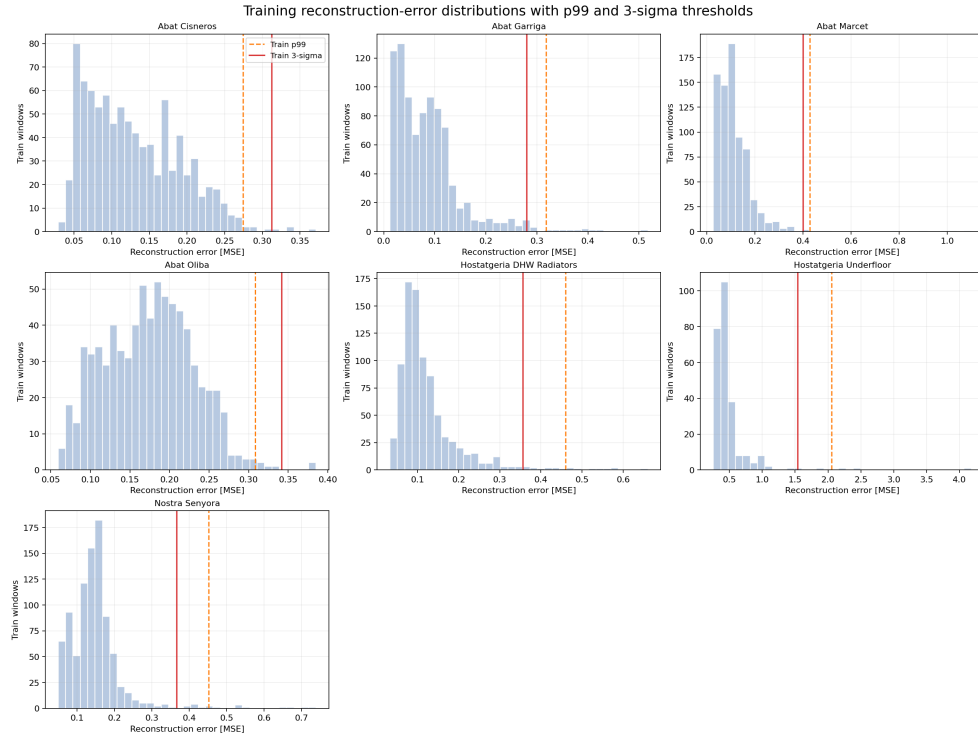


Figure 2: Training reconstruction-error distributions for the modeled sheets, shown with both the training 99th percentile threshold and the 3-sigma threshold. The difference between the two thresholds depends on the shape and spread of the training reconstruction-error distribution.

Table 1: Five-building result summary under the 3-sigma threshold.

Building	Windows	Flagged	Rate [%]	Top feature
Abat Cisneros	1113	4	0.36	Flow
Abat Garriga	1107	17	1.54	Return
Abat Marcet	951	10	1.05	Flow
Abat Oliba	955	2	0.21	Return
Hostatgeria Underfloor	323	5	1.55	Supply

5 RESULTS

5.1 Current main result set

The current end-to-end result set used for the draft focuses on five heating-consumer sheets: Abat Cisneros, Abat Garriga, Abat Marcet, Abat Oliba, and Hostatgeria Underfloor. These five buildings were kept because they provided the most usable retained windows without the long faulty constant-value periods observed in two of the exploratory cases. The strongest anomaly case remains `cons_hostatgeria_underfloor_he`, while the five-building set as a whole still shows substantial variation in anomaly count, dominant feature, and engineering-baseline overlap.

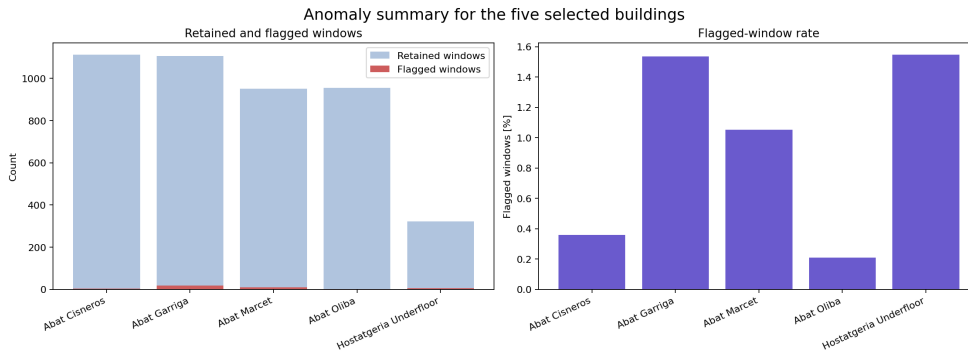


Figure 3: Retained-window count, flagged-window count, and flagged-window rate for the five selected buildings.

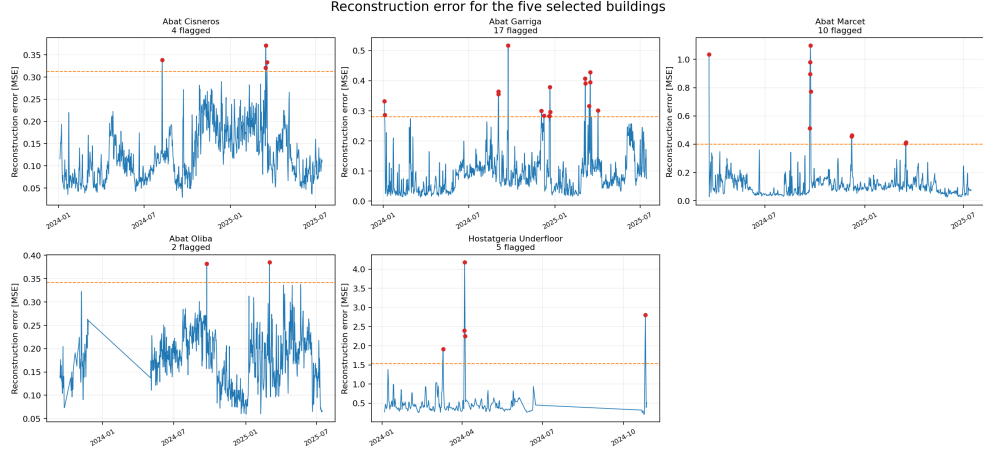


Figure 4: Reconstruction-error timelines for the five selected buildings. The horizontal dashed line is the anomaly threshold and red points are flagged windows.

5.2 Dominant-feature interpretation

The clearest current anomaly-typing view is the dominant-feature analysis. Across the five-building set, the anomalies are mainly flow-dominant and return-dominant, with Hostatgeria Underfloor providing the clearest supply-dominant case. This indicates that the autoencoder is detecting multiple anomaly families rather than only one repeated signature.

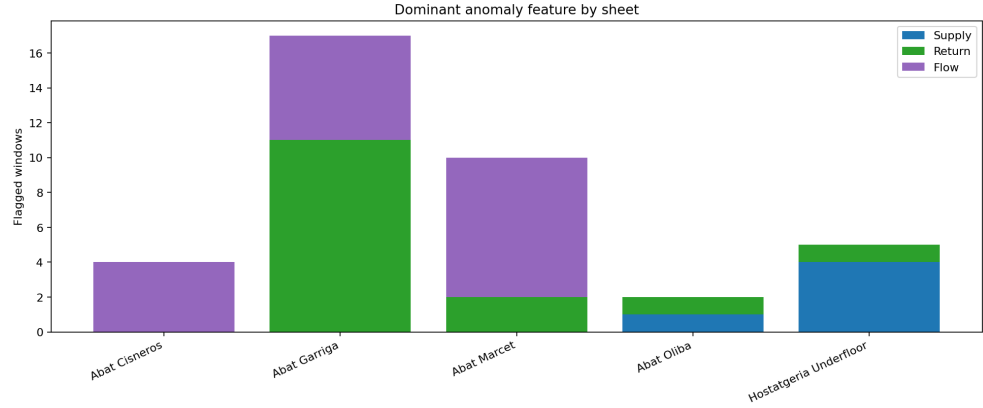


Figure 5: Dominant anomaly feature by sheet for the five selected buildings.

5.3 Low delta-T overlap

Only one sheet, `cons_hostatgeria_underfloor_he`, shows strong overlap between top reviewed autoencoder anomalies and the low delta-T baseline. This should be interpreted carefully. It makes underfloor heating the strongest validated case, but the absence of overlap in the other four buildings does not invalidate the autoencoder. It instead suggests that many detected anomalies are not low delta-T events and therefore require interpretation through dominant feature, flow-channel behavior, and domain review.

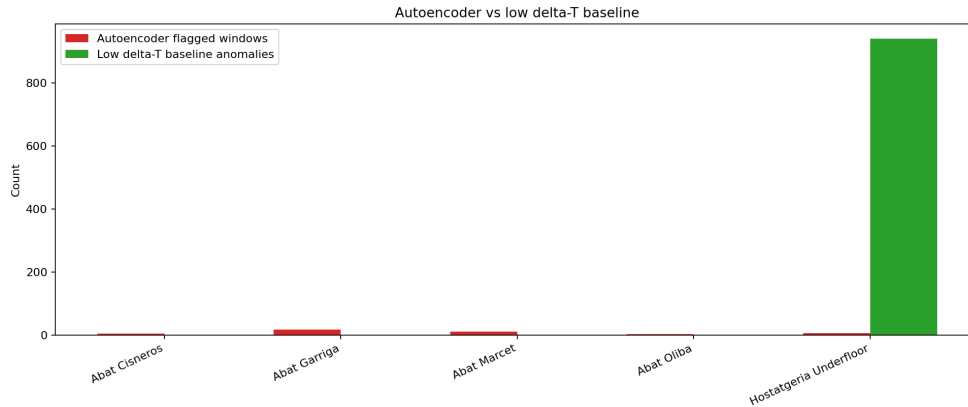


Figure 6: Autoencoder anomaly counts compared with the engineering low delta-T baseline for the five selected buildings.

5.4 Operating regimes

Operating-regime clustering indicates that daily windows do not form one undifferentiated population. Some sheets occupy very distinctive regimes, and regime occupancy is uneven. Underfloor heating stands out as the most distinctive operating-regime case. This supports the conclusion that its anomalies are associated with a coherent subsystem behavior rather than random variation.

5.5 Threshold comparison

The comparison between p99 and 3-sigma shows that threshold choice materially affects anomaly counts. The effect is sheet-dependent rather than global: some sheets become more selective under 3-sigma, while others become less selective. This is a useful result in itself because it shows that the training-error distribution shape must be considered when choosing an anomaly threshold.

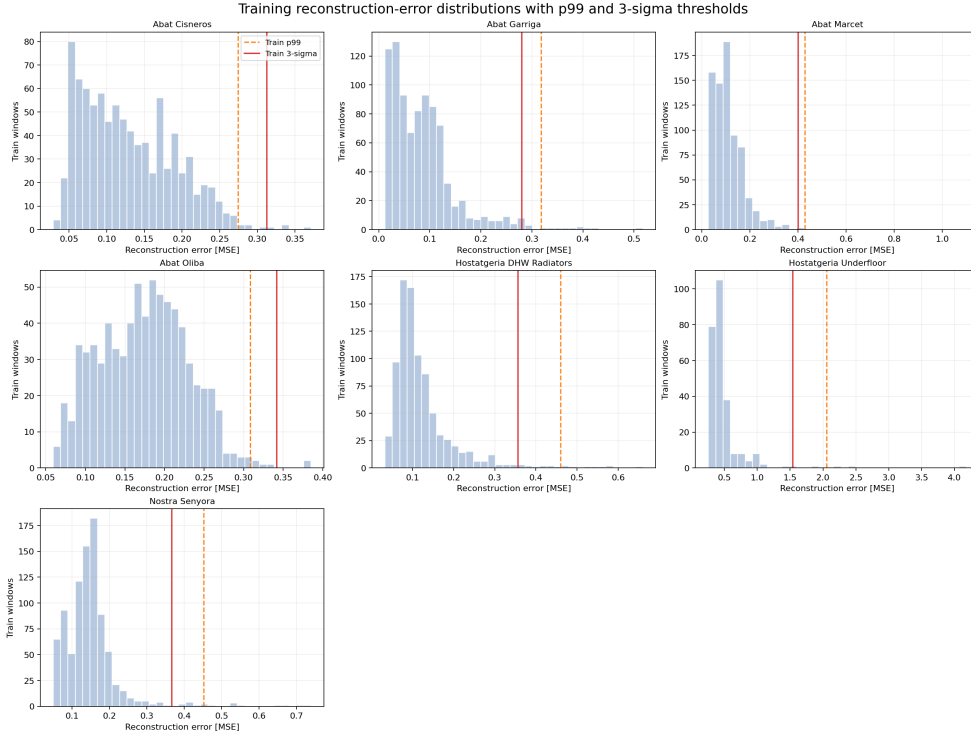


Figure 7: Training reconstruction-error distributions with the p99 and 3-sigma thresholds overlaid.

5.6 Flow-channel sensitivity comparison

During model development, the anomaly counts were found to be sensitive to how the flow channel was represented. The clearest example is `cons_abat_oliba`, where the direct raw-flow version produced windows dominated by very small delta-T and inflated flow values, while the final flow-channel representation yielded anomalies that were more interpretable in terms of supply or return behavior.

5.7 Train-test split and anomaly timing

The anomalies are not confined to the first chronological portion of the data. Some buildings show both training and later test-period anomalies, which is useful when thinking about eventual live deployment.

6 DISCUSSION

6.1 What is currently convincing

The most convincing current conclusion is that reconstruction-based anomaly detection is viable for this dataset and that `cons_hostatgeria_underfloor_he` provides a strong thesis case. It is physically distinctive, it overlaps the engineering baseline, and its anomaly patterns remain interpretable under both raw and stabilized flow views.

6.2 What remains uncertain

The biggest remaining limitation is still validation. Most sheets do not have labeled faults, so anomaly interpretation depends on triangulation:

- reconstruction error
- dominant feature
- signal plots
- low delta-T overlap
- supervisor and domain review

This means the thesis should frame most anomalies as credible candidates rather than confirmed faults.

6.3 Implications for live deployment

The historical result pack is already useful for deployment preparation. Once one to two months of live data become available, the intended strategy is to apply the historical-trained detector prospectively and evaluate the flagged windows through operational review rather than standard classification metrics.

6.4 Writing priorities

When drafting the final thesis, the narrative should remain focused. The autoencoder is the main method. The low delta-T baseline is the main engineering comparison. Dominant-feature attribution is the main anomaly interpretation layer. Clustering should be presented as a supporting analysis rather than the core detection method.

7 CONCLUSION

This thesis develops a reconstruction-based anomaly-detection pipeline for a small district-heating network with limited fault labels. The work adapts ideas from the HEAT paper to a setting where long historical coverage is more useful than large-scale peer-group clustering. The current results show that the method can identify interpretable anomalous operating windows and that the strongest case is underfloor heating, where the detected anomalies also align with a simple engineering low delta-T rule.

The broader seven-sheet analysis shows that anomaly behavior is heterogeneous across the network. Flow-dominant anomalies are the most common, return-dominant anomalies are the second main group, and supply-dominant anomalies are less frequent. Stabilized flow improves interpretability relative to raw flow and is therefore the main reporting lens. Operating-regime clustering remains useful for context, but dominant-feature attribution is the more effective explanation layer.

The next stage of the work is to apply the current detector to live data and use that prospective period as an operational evaluation step. This will help determine whether the anomaly candidates identified in historical data correspond to meaningful ongoing system behavior.

ACKNOWLEDGMENTS

The acknowledgments section should be finalized once the supervisor, collaborators, and external contributors to the project are confirmed.

REFERENCES

A WRITING NOTES

This appendix can be replaced later with:

- additional figures
- threshold comparison tables
- supervisor review artifacts

- extended anomaly examples